

US NATURAL DISASTER TEMPORAL ANALYSIS & VISUALIZATION

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US Natural Disaster Temporal Analysis & Visualization

Milestone - 3

1. Project Description

The US Natural Disaster Geospatial Analysis project focuses on analyzing and visualizing the geographic distribution of disaster declarations across the United States. This milestone expands the earlier temporal analysis by introducing geospatial exploratory data analysis (EDA) to identify regional patterns and high-risk zones.

Key Objectives:

- Spatial Distribution: Map disaster frequencies across US states.
- Regional Specialization: Identify which disaster types dominate in specific regions.
- Hotspot Detection: Detect geographic clusters for events such as hurricanes and floods.
- Geographic Insights: Explain why some states experience more disaster declarations.

2. Dataset Used

The project uses the Federal Emergencies and Disasters dataset (1953–Present). The dataset contains historical records of disaster declarations in the United States and is used to analyze spatial patterns of disasters across different states.

Dataset Characteristics:

- Geographic Data: Contains state identifiers used for mapping.
- Categorical Data: Includes incidentType such as Fire, Flood, and Hurricane.
- Time Features: Enables extraction of year and month for seasonal analysis.
- Cleaned Dataset: Uses the cleaned dataset from Milestone 1 and 2.

3. Environment Setup

The analysis is implemented using Python in Visual Studio Code with Jupyter Notebook for execution and visualization.

Tools and Libraries Used:

- Pandas & NumPy – Data loading and preprocessing.
- Matplotlib & Seaborn – Static visualizations.
- Plotly – Interactive geographic maps.
- GeoPandas – Handling geographic data.

- Folium – Interactive hotspot maps.
- Jupyter Notebook – Cell-based workflow for analysis.

4. Data Exploration & Validation

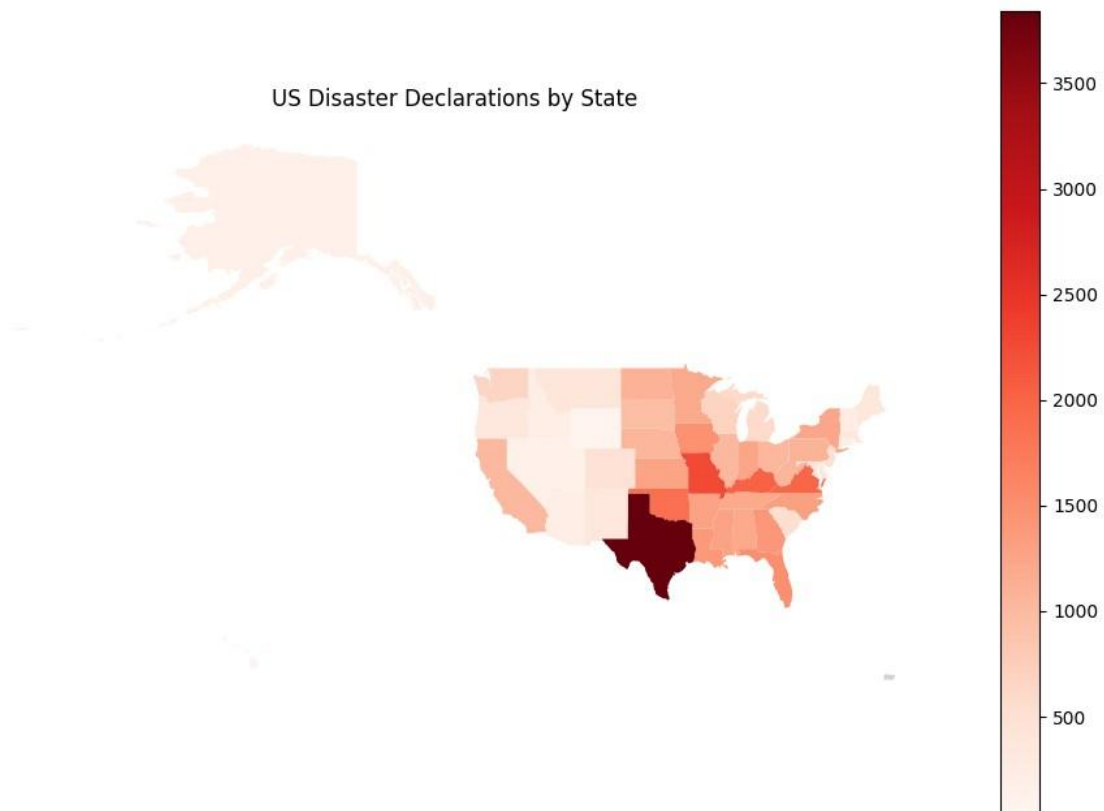
Before creating geospatial visualizations, the dataset was validated to ensure that it contains the required geographic information. **5. Geospatial Trend Analysis**

5.1 Nationwide Distribution (Plotly Choropleth)

A choropleth map visualizes the total number of disaster declarations per state. Darker colours represent states with higher disaster frequencies.

Figure 1: Nationwide Disaster Distribution Map

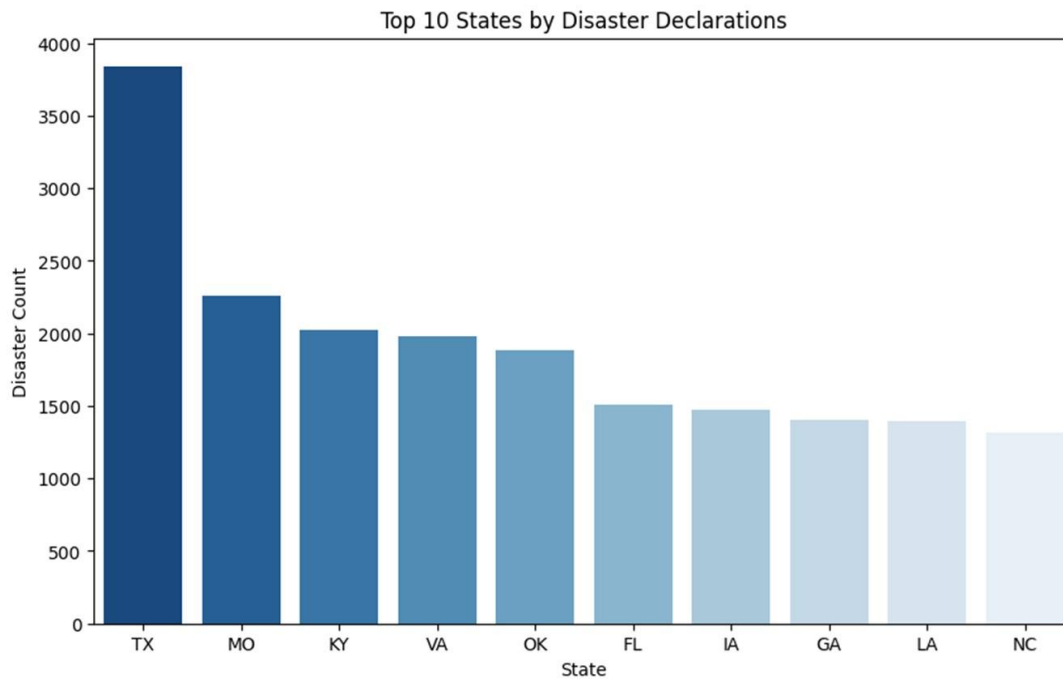
Choropleth Map



5.2 Top 10 State Comparative Analysis (Seaborn)

A bar chart compares the top ten states with the highest number of disaster declarations.

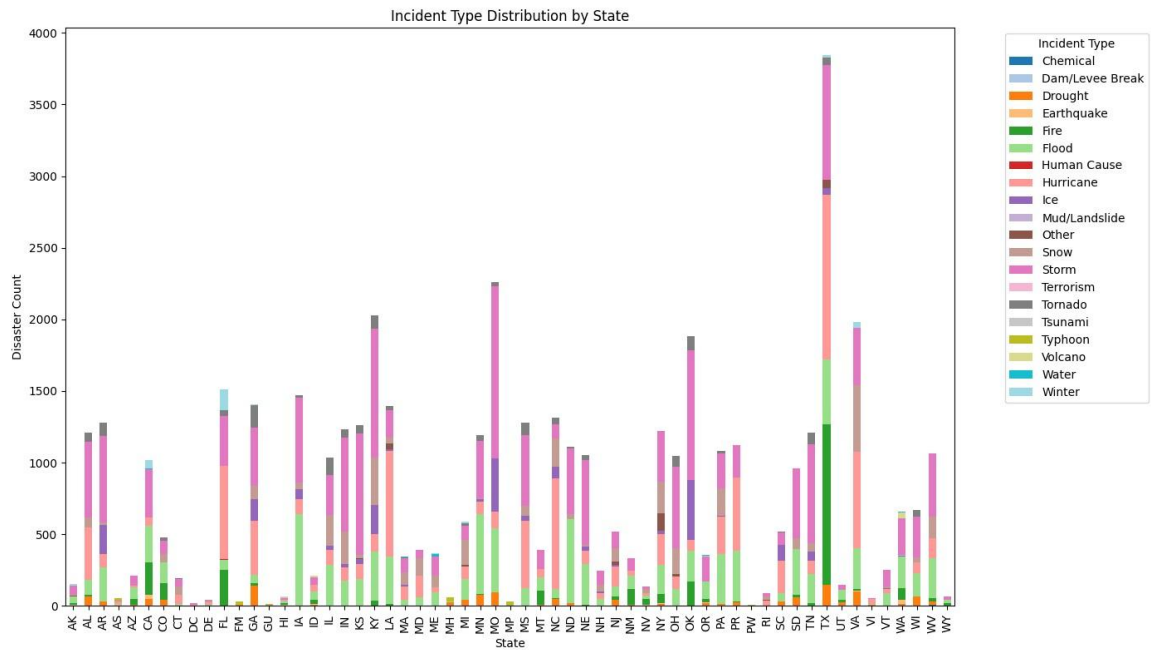
Figure 2: Top 10 States by Disaster Declarations



5.3 Geographic Incident Composition (Stacked Bar)

A stacked bar chart shows the distribution of disaster types within the top 10 states, revealing each state's disaster signature.

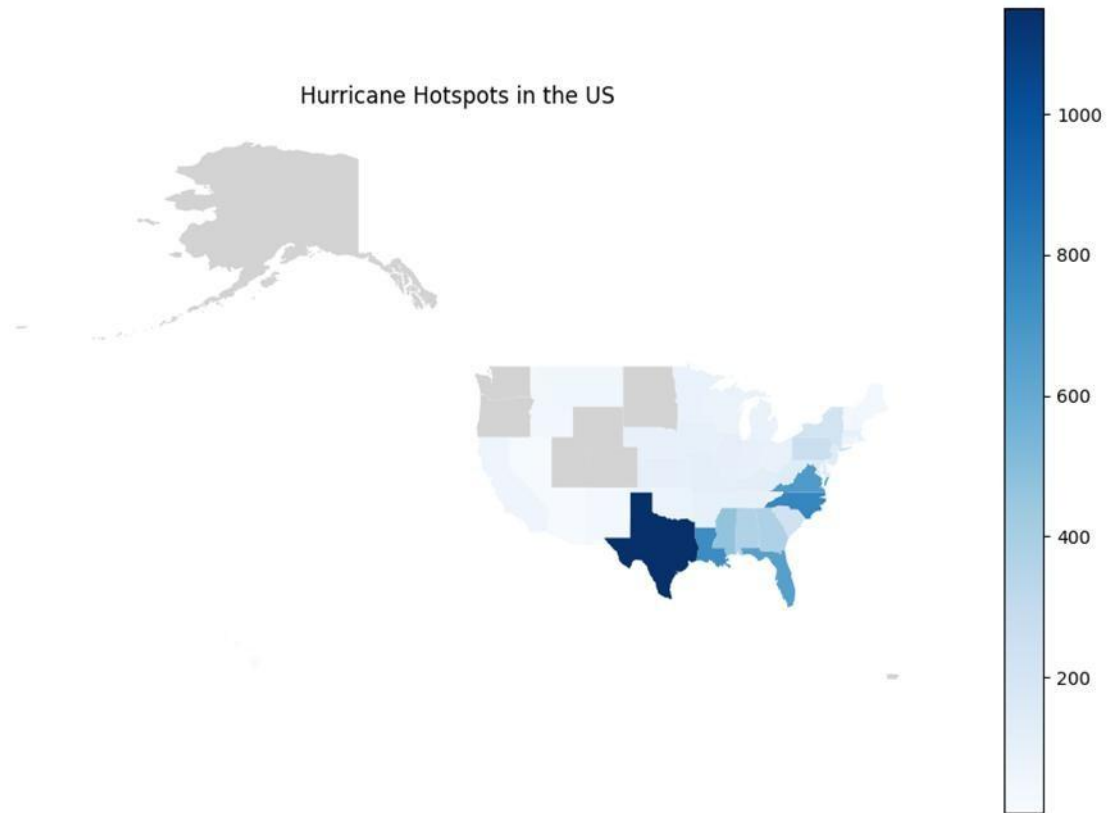
Figure 3: Disaster Type Composition by State



5.4 Interactive Hurricane Hotspot Detection (Folium)

An interactive map highlights hurricane-prone regions primarily located along the Gulf Coast and southeastern United States.

Figure 4: Hurricane Hotspot Map



6. Visualization Standards

maintain clarity and consistency, the following visualization standards were applied:

- Clear legends explaining colour intensity.
- Proper axis labels and chart titles.
- Reduced visual clutter and readable font sizes.
- Interactive features for map exploration.

7. Conclusion

This milestone successfully analyzed the geographic distribution of disaster declarations across the United States. The visualizations highlight regional risk patterns and demonstrate how different disaster types are concentrated in specific geographic areas.